$$P(\overset{Y}{H} \mid x) = P(buy = y \mid \hat{x}) = P(\hat{x} \mid buy = y)\, \cancel{P(buy = yes)}\ \tfrac{9}{14}$$

$$\cancel{P(age \le 30 \mid yes)}\ \tfrac{2}{9}\ \cancel{P(income = med \mid yes)}\ \tfrac{4}{9}\ P(student \mid yes)\ \tfrac{6}{9}\ \cancel{P(Credit = fair \mid yes)}\ \tfrac{6}{9}$$

$$= \left(\frac{2}{9}\right)\left(\frac{4}{9}\right)\left(\frac{6}{9}\right)\left(\frac{\cancel{6}^{3}}{\cancel{9}}\right)\left(\frac{\cancel{9}^{1}}{\cancel{14}_{7}}\right)$$

$$= \frac{16}{567} = 0.028$$

$$P(buy = N \mid \hat{x}) = \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5} \times \frac{5}{14} = 0.07$$

Quiz 12

♀ Age 20 Stu yes

$$P(buy = y \mid \hat{x}) = P(\hat{x} \mid buy = y)\, \cancel{P(buy = y)}\ \tfrac{9}{14}$$

$$= \cancel{P(age \le 30 \mid yes)}\ \tfrac{2}{9}\ \cancel{P(student \mid yes)}\ \tfrac{6}{9}\ \cancel{P(buy = y)}\ \tfrac{9}{14}$$

$$= \left(\frac{2}{9}\right)\left(\frac{6}{9}\right)\left(\frac{9}{14}\right) = \frac{2}{21} = 0.095$$

$$P(buy = N \mid \hat{x}) = P(\hat{x} \mid buy = N)\ P(buy = N)$$

$$= P(age \le 30 \mid no)\ P(student \mid no)\ P(buy = N)$$

$$= \left(\frac{3}{5}\right)\left(\frac{1}{5}\right)\left(\frac{5}{14}\right) = 0.042$$

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